

**MULTIBAND MICROSTRIP ANTENNA-BASED
BREAST TUMOR
DETECTION USING MACHINE LEARNING
CLASSIFICATION**

An Internship Report

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by

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CERTIFICATE

This is to certify that Mr. Shivam Kumar, a student of B.Tech (ECE), 3rd Year, Silicon Institute of Technology, Sambalpur, has successfully completed his summer internship in the Department of Electronics and Communication Engineering, National Institute of Technology, Jamshedpur.

During the internship, he worked on the project entitled “Multiband Microstrip Antenna-Based Breast Tumor Detection Using Machine Learning ” under my guidance.

The work presented in this report was carried out by him during the internship period. To the best of my knowledge, this report is an original work completed by the student under my supervision.

I wish him all the best for his future studies and career.

Prof. Surajit Kundu

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DECLARATION

I, Shivam Kumar, a B.Tech (3rd Year) student of the Department of Electronics and Communication Engineering, Silicon Institute of Technology, Sambalpur, hereby declare that the work presented in this report, titled “*Multiband Microstrip Antenna-Based Breast Tumor Detection Using Machine Learning Classification*”, was carried out by me during my summer internship at the Department of Electronics and Communication Engineering, National Institute of Technology, Jamshedpur, under the guidance of Prof. Surajit Kundu.

I declare that the antenna design, breast phantom simulations, dataset preparation, and machine learning analysis presented in this report were carried out as part of my internship work. This report has not been submitted to any other institute for the award of any degree.

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Shivam Kumar

ABSTRACT

Among the various cancers that exist in women, breast cancer is considered to be one of the most common types. Different techniques could be employed in order to detect the presence of breast cancer in patients. However, some weaknesses can be identified in the process of using those techniques to detect the presence of breast cancer which include: high cost, radiation (use of X-ray), discomfort, and unavailability. The current research paper deals with the development of a microwave antenna for detecting breast tumor. Design and simulation of microstrip patch antenna was carried out via CST Studio Suite. Also, the creation of 3D breast phantom with placement of tumor at different locations was done. Data obtained from antenna was analyzed especially S11 data of antenna with and without the presence of tumor. Data simulated from CST was used to generate datasets and classification of those was done by means of machine learning algorithm. Various machine learning algorithms were compared in terms of ability to classify samples into tumor/no tumor. It is found that machine learning could be used in order to classify samples for detecting the breast tumor based on the data obtained from the simulation of antenna. This research was conducted as a part of internship in the National Institute of Technology, Jamshedpur.

A compact $50\text{ mm} \times 50\text{ mm}$ microstrip patch antenna was designed using an FR4-lossy substrate with relative permittivity of $\epsilon_r = 4.3$. Simulation of the antenna was done via CST Studio Suite over the frequency range of 0.3–5 GHz. Four resonant frequencies could be identified at 1.05, 1.74, 2.63, and 4.65 GHz. The best return loss value of -35.60 dB was obtained, along with VSWR of 1.03 at 2.63 GHz. The gain of the antenna was estimated as 2.08 dBi. Model of a half-sphere in 3D for the breast phantom was created via CST software after simulation of the antenna with the presence of layers of skin, fat, glandular, and tumor tissue layers. Tumors with sizes 5mm, 10mm, and 15mm were placed in different locations within breast phantom structure. Altogether there were 30 tumors, as well as 5 situations without any tumor, giving 35 sets of S11 values. Ten features were extracted from each S11 graph and used in machine learning classifiers. Eight machine learning classifiers were used: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, Random Forest, XGBoost, CatBoost, and Artificial Neural Network (ANN). Ratio between training and testing data was processed, especially S11 data of antenna with presence or absence of tumor. The data simulated from CST was used to develop datasets and these were further classified with the help of machine learning algorithm. Various machine learning algorithms were compared to classify samples for tumor or no tumor. It was found that machine learning can be used by taking simulated antenna data for detecting breast tumor. This research work was done during the period of summer internship at the National Institute of Technology, Jamshedpur.

A compact 50 mm × 50 mm microstrip patch antenna was designed using an FR4-lossy substrate with a relative permittivity of $\epsilon_r = 4.3$. The antenna was simulated in CST Studio Suite over a frequency range of 0.3–5 GHz. Four main resonant frequencies were observed at 1.05, 1.74, 2.63, and 4.65 GHz. The best return loss obtained was -35.60 dB, with a VSWR of 1.03 at 2.63 GHz. The simulated gain of the antenna was approximately 2.08 dBi. The model of the half-sphere in 3D for the breast was constructed through the CST software after the simulation of the antenna in presence of skin, fat, glandular, and tumor tissue layers. The tumors of the sizes 5mm, 10mm, and 15mm were located in different positions in the breast structure. Overall, there were 30 tumors, as well as five scenarios without tumors, making 35 sets of S11. Ten features have been extracted from each S11 graph and used in machine learning classifiers. Eight machine learning classifiers have been utilized: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, Random Forest, XGBoost, CatBoost, and Artificial Neural Network (ANN). Split between training and testing data was set to 80/20 ratio. KNN, Decision Tree, and CatBoost yielded 100% accuracy on testing data set. Of those classifiers, KNN is chosen to be the best one for this study because of its simplicity and fast training time. This research demonstrates that using microwave antenna sensing together with machine learning approaches can be useful in recognizing the difference between tumors and non-tumor samples. Nevertheless, more data and testing are necessary for practical usage.

Keywords: *Breast Tumor Detection; Microstrip Patch Antenna; Microwave Sensing; CST Studio Suite; Breast Phantom; Machine Learning; K-Nearest Neighbours; S11 Classification.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CST	Computer Simulation Technology
FR4	Flame Retardant 4
S11	Input Reflection Coefficient
VSWR	Voltage Standing Wave Ratio
dB _i	Decibels relative to an isotropic radiator
ML	Machine Learning
KNN	K-Nearest Neighbours
SVM	Support Vector Machine
ANN	Artificial Neural Network
XGBoost	Extreme Gradient Boosting
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
ϵ_r	Relative Permittivity
σ	Electrical Conductivity
μ_r	Relative Permeability

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Breast cancer is one of the most common types of cancer found in women around the world. It is also one of the major causes of cancer-related deaths among women. According to the World Health Organization (WHO), millions of women are affected by breast cancer every year. The number of new cases is also increasing in many countries [1]. More than two million new cases are reported globally each year, which shows that breast cancer is a major health problem [2].

The stage of breast cancer at which it is diagnosed plays a critical role in determining survival chances. When the diagnosis is done at an early stage while the cancerous cells are confined within the breast, treatment tends to be more effective. Early diagnosis may also lower the requirement for aggressive therapy. For this reason, numerous scientists are attempting to develop means of early detection of breast cancer. The tests developed ought to be accurate, cost-effective, and user-friendly in areas with no advanced imaging technologies.

This work deals with the detection of breast cancer through microwave sensing. Machine learning algorithms have been used to classify and compare the simulated data. This paper highlights the antenna design, electromagnetic simulations, phantom modeling of the breast with tumor, and machine learning algorithms performed as part of my internship at NIT Jamshedpur.

1.2 Limitations of Existing Breast Cancer Detection Methods

X-ray mammography, MRI, and ultrasound are commonly used for breast cancer detection [3]. However, each technology has limitations. First, mammography involves exposure to X-rays and requires compression of the breast, which can be painful. Second, it can produce incorrect diagnoses both positive and negative. Ultrasound helps visualize the tissues of the breast; however, it may provide images of lesser quality compared to mammography in certain situations. MRI produces high-quality images; however, it is expensive and requires special equipment.

These limitations have encouraged researchers to study other methods for breast cancer detection. Microwave-based detection is one such method that can be explored along with antennas and machine learning. [3].

MRI imaging is effective in the detection of breast lesions, particularly when dealing with women who have dense breasts. However, MRI imaging is expensive, time-consuming, and might require a contrast medium. These factors make it unsuitable for breast screening. Ultrasound imaging doesn't utilize any radiation and is generally less expensive than MR imaging. However, the outcome will highly rely on the skills of the operator.

These limitations show the need for a better screening method that does not use harmful radiation, is affordable, easy to carry, and can detect the changes in the electrical properties of breast tissue caused by tumor growth.

1.3 Microwave Sensing for Breast Tumor Detection

Microwave sensing is based on the difference in electrical properties between normal and tumor breast tissues. Tumor tissue generally has higher dielectric properties, such as relative permittivity and conductivity, compared to surrounding normal tissue. Lazebnik et al. studied the dielectric properties of normal, benign, and malignant breast tissues over a wide range of microwave frequencies and found a clear difference between them [4]. This difference makes it possible to detect a tumor using a microwave antenna. When the antenna is placed near the breast, the presence of a tumor can change the reflected signal of the antenna. This change can be measured using parameters such as S_{11} .

Microwave sensing uses non-ionising radiation and can operate at relatively low power levels. Therefore, it does not have the same radiation concerns as X-ray-based mammography. The basic setup can use a small antenna along with a vector network analyser (VNA) or a similar measurement system. This makes microwave sensing a possible low-cost and portable method for breast tumor detection. Many researchers have studied microstrip patch and ultra-wideband antennas for this purpose, which are discussed further in Chapter 2 [5, 6].

1.4 Role of Machine Learning in Tumor Classification

Data collected from the antenna such as the reflection coefficient (S_{11}) and the transmission coefficient (S_{21}) on their own will not be able to provide information about whether there is a tumor present. There is a possibility that the change in the S_{11} value may have occurred due to a tumor but it is also possible that the change may be due to changes in the tissue environment as well as due to noise in the measurements.

Machine learning can be used to analyse the antenna data and make the classification process easier. In supervised learning, the model is trained using labelled samples, such as tumor-present and tumor-absent cases. The model learns the patterns in the input data and then uses them to classify new samples. This can be more useful than using a fixed threshold on the S_{11} values. Machine learning has been used in different breast cancer detection tasks, including image analysis and clinical data classification [7, 8]. The use

of machine learning with antenna-based sensing is also being studied for detecting breast tumors from S-parameter data [9, 10].

In this work, the combination of the antenna simulation and the machine learning algorithm will be used to identify the presence of breast cancer. Simulations of the antenna along with a breast phantom are carried out using CST software for the existence of the cancerous growth and absence of the cancerous growth, and the S11 values are obtained. These values will then be fed into different machine learning algorithms, and the one with better prediction accuracy will be determined.

1.5 Objectives of the Internship

The internship was carried out with the following main objectives:

1. To design and simulate a compact microstrip patch antenna for breast tumor detection and study its S11, bandwidth, VSWR, gain, and radiation pattern.
2. To create a three-dimensional breast phantom in CST Studio Suite using different tissue layers such as skin, fat, and glandular tissue.
3. To place tumors at different positions in the breast phantom and simulate the antenna response for both tumor and tumor-free cases.
4. To collect the S11 data from the different simulation cases and prepare a labelled dataset for machine learning.
5. To apply different machine learning algorithms to the collected data and compare their performance for classifying tumor and non-tumor cases.
6. To find the best-performing machine learning model for the generated dataset and study its results and limitations.

1.6 Organisation of the Report

The rest of this report is organised into different chapters. Chapter 2 presents the literature review related to microwave-based breast tumor detection, antenna design, breast tissue properties, and the use of machine learning in cancer detection. Chapter 3 explains the basic theory related to the antenna, S-parameters, breast tissue, and machine learning. Chapter 4 describes the proposed methodology, including the antenna design, breast phantom model, tumor placement, CST simulation, and dataset generation. Chapter 5 presents the antenna design and CST simulation results, including S11, bandwidth, VSWR, gain, and radiation pattern. Chapter 6 discusses the breast phantom simulations and the generated dataset. Chapter 7 explains the machine learning methods used for classification. Chapter 8 presents and discusses the results obtained from the different machine learning models. Finally, Chapter 9 gives the conclusion, limitations, and future scope of the work.

CHAPTER 2

LITERATURE REVIEW

This chapter discusses earlier research done on the detection of breast tumors using microwaves. The areas of discussion include antenna design, characteristics of breast tissue, breast phantom modeling, and machine learning. Various research works analyze the electrical characteristics of normal tissue and tumor to detect the tumor. Different antenna design, simulation techniques, and machine learning techniques have been applied for this purpose. This chapter focuses on some important findings of the past research works and identifies the research gap for the current research.

2.1 Microwave Imaging Techniques for Breast Cancer Detection

Microwave techniques for breast tumors identification have been widely researched as either replacement or complement to the standard mammography techniques. The concept is that the tumor tissue will have unique electrical properties compared to the healthy breast tissue at microwave frequencies. There are several techniques to exploit this difference. There are those that will generate images based on microwave signals generated within the tissue; there are also those that analyze the reflected microwave signals. A more straightforward technique would be to analyze the reflection coefficient of the antenna.

Mahfuz et al. reviewed different microstrip patch antenna-based methods used for breast tumor detection. Their study discusses different antenna shapes, substrate materials, and operating frequency ranges used by researchers [5]. Saeidi et al. reviewed ultra-wideband (UWB) antennas used for microwave-based breast tumor detection and discussed how wider bandwidth can help in detecting and locating tumors more accurately [6].

The technique utilizes a single antenna and monitors the response of S_{11} rather than creating the image of the breast. It helps to make the process simple and reduces the cost of hardware involved. However, it does not provide the location of the tumor in the breast since this is done by imaging devices. In order to optimize the S_{11} data, machine learning classifies the tumor from the signal of the antenna. This use of machine learning is discussed further in Section 2.5.

2.2 Antenna Design for Biomedical Sensing Applications

Microstrip patch antennas are widely used for breast tumor sensing because they are small, lightweight, easy to design, and can be fabricated at a relatively low cost. They are also suitable for applications where the antenna needs to be placed close to the body. Çalışkan et

al. designed a rectangular microstrip patch antenna using an FR4 substrate with an operating frequency close to 2.45 GHz for breast cancer detection. Their results showed that the resonant response of the antenna changed when a tumor-like object was placed inside the breast model [3]. Sarker et al. studied a meta material-loaded antenna array for breast tumor detection. Their work showed that artificial magnetic conductor (AMC) loading can improve antenna gain and front-to-back ratio, which can be useful when the antenna is placed near biological tissue [11].

One common idea found in many of these studies is to use multiband or ultra-wideband antennas. A wider frequency range allows the antenna to interact with breast tissue at different frequencies. Different frequencies can respond differently to tissue properties and tumor sizes. Using more than one frequency can therefore provide more information about the presence of a tumor compared to using only one frequency. The antenna designed in this project also follows this approach and shows four different resonant frequencies within the 0.3–5 GHz range.

2.3 Dielectric Properties of Breast Tissue and Phantom Modelling

The dielectric properties of breast tissue are important when simulating a microwave-based breast tumor detection system. These properties include relative permittivity and electrical conductivity, which affect how microwave signals travel through different tissues. Lazebnik et al. measured the microwave dielectric properties of normal, benign, and malignant breast tissues and found that tumor tissue generally has higher permittivity and conductivity than normal breast tissue [4]. These measured values are useful for assigning suitable material properties to breast tissue models used in simulation. Similar values were also considered while creating the breast phantom used in this project.

Simulation models usually have only a few layers that are associated with breast tissue types including skin, fat, and glandular tissues. The tumor is introduced in one region of this model as an inclusion having various dielectric properties inside a breast with various sizes. A hemispherical or similar structure may be used as the overall geometry of the breast. Such models simplify the simulations while providing a good description of breast tissues. A similar layered phantom model was used in this project for the CST simulations [3, 10].

2.4 Machine Learning in Medical Diagnostics

Apart from antenna-based methods, machine learning is also widely used in breast cancer detection. It is mainly used to classify medical or image-based data into different categories, such as benign and malignant cases. Dinesh et al. compared five machine learning algorithms, namely Support Vector Machine (SVM), KNN, Logistic Regression, Random Forest, and Decision Tree, using a breast cancer image-derived feature dataset. Their study showed that the choice of the machine learning algorithm can affect the accuracy of breast

cancer classification. Different algorithms gave different performance on the same type of data [7].

Uddin et al. compared different supervised machine learning algorithms on several disease prediction datasets. They found that no single machine learning algorithm always gives the best result. The performance of a model depends on factors such as the size of the dataset, the quality of the features, and the balance between the classes [8]. This is also important in the present project because the dataset used here is small and contains 30 tumor-present samples and 5 tumor-absent samples. Austria et al. studied LASSO-based feature selection along with different classifiers for breast cancer detection. Their work shows that proper preprocessing and selection of useful features are important before training a machine learning model [12].

In this project, different machine learning algorithms were also tested, including KNN, Support Vector Machine, Random Forest, XGBoost, and CatBoost. These algorithms are based on well-known machine learning methods developed through earlier research. The KNN method was studied by Cover and Hart [13], while the Support Vector Machine method was introduced by Cortes and Vapnik [14]. Random Forest was proposed by Breiman [15]. XGBoost and CatBoost are gradient-boosting methods developed by Chen and Guestrin [16] and Prokhorenkova et al. [17], respectively. An Artificial Neural Network was also evaluated in this project. The working and implementation of these machine learning algorithms are explained in more detail in Chapter 5.

2.5 Related Work Combining Antenna Sensing and Machine Learning

This combination of antenna-based sensing and machine learning techniques has been used in the detection of breast cancer tumors. The idea is to use data obtained from the antenna, such as S-parameter measurements, to train a machine learning algorithm to classify the data. This is unlike techniques that generate an entire microwave image of the breast.

Alharbi et al. studied a machine learning-based ultra-wideband microstrip antenna system for microwave imaging applications. In their work, the antenna was used to obtain the required microwave response, and machine learning was then used to identify unusual scattering responses [9]. Rana et al. used radar cross-section (SAR) data obtained from a Ultra Wide Band (UWB) monopole antenna along with a convolutional neural network (CNN) to detect malignant tissue. Their study showed that machine learning can learn useful features from the antenna data when enough training data is available [10].

All these above studies indicate the feasibility of combining the use of machine learning and antenna sensing for detection of tumors. Instead of forming the full image, antenna response is used as an input for machine learning classifier. In the same way, this current work uses the data from S-parameters received from CST simulation results as input for various machine learning algorithms.

2.6 Research Gap and Motivation for the Present Work

The previous studies show that microwave antennas can be used for breast tumor detection and that machine learning can help in classifying the obtained data. However, many existing studies mainly focus on only one part of the problem. Some studies mainly focus on antenna design and its electromagnetic performance, without using machine learning for further classification. Other studies use machine learning on existing medical or clinical datasets that are not directly connected to a particular antenna design.

There are comparatively fewer studies that combine all the steps together, starting from antenna design and continuing up to machine learning-based classification. In this type of approach, the antenna is designed first, a breast phantom is created with different tumor conditions, the antenna response is obtained for each condition, and then the data is used to train and compare different machine learning models.

This project follows this complete process. First, a multiband microstrip patch antenna is designed and its performance is studied, as explained in Chapter 3. Then, a hemispherical breast phantom is created with different tumor sizes of 5 mm, 10 mm, and 15 mm, as discussed in Chapter 4. The S11 responses obtained from the simulations are then used to extract ten spectral features. These features are given to eight different machine learning classifiers for training and comparison. The machine learning methods used in the project include different types of models such as KNN, Support Vector Machine, Random Forest, XGBoost, CatBoost, and Artificial Neural Network were used. Their results are discussed in Chapters 5 and 6. Thus, this project tries to connect antenna design, breast phantom simulation, S11 feature extraction, and machine learning in a single workflow.

CHAPTER 3

ANTENNA DESIGN AND SIMULATION

This chapter highlights the design and analysis of the proposed microstrip patch antenna using CST Studio Suite. Performance of the antenna is analyzed based on parameters like return loss, impedance bandwidth, VSWR, and far field radiation pattern.

3.1 Antenna Design Specifications

The antenna used in this project is a microstrip patch antenna. It was selected because it has a simple and flat structure, is lightweight, and can be easily used for breast tumor sensing. The antenna is fed using a microstrip line. The radiator, feed line, and ground are all placed on the same FR4 substrate. The main design specifications of the antenna are given in Table 3.1 below.

Table 3.1: Antenna design specifications

Parameter	Value
Substrate material	FR-4 (lossy)
Relative permittivity, ϵ_r	4.3
Substrate footprint	50 mm $\times$ 50 mm
Antenna type	Microstrip patch (planar)
Feed mechanism	Microstrip line feed
Simulation software	CST Studio Suite
Simulated frequency range	0.3 GHz – 5 GHz
Simulation domain	Time-domain solver, free space boundary

The dimensions of the patch and feed line were selected by changing the antenna dimensions in CST and checking the results each time to find the dimensions that gave better performance. This process was repeated until the required multiband response was obtained in the 0.3–5 GHz frequency range, as discussed in Section 3.3. Figure 3.1 shows the antenna model in CST. The antenna is placed facing the breast phantom used for the tumor-sensing simulations, which is explained in detail in Chapter 4.

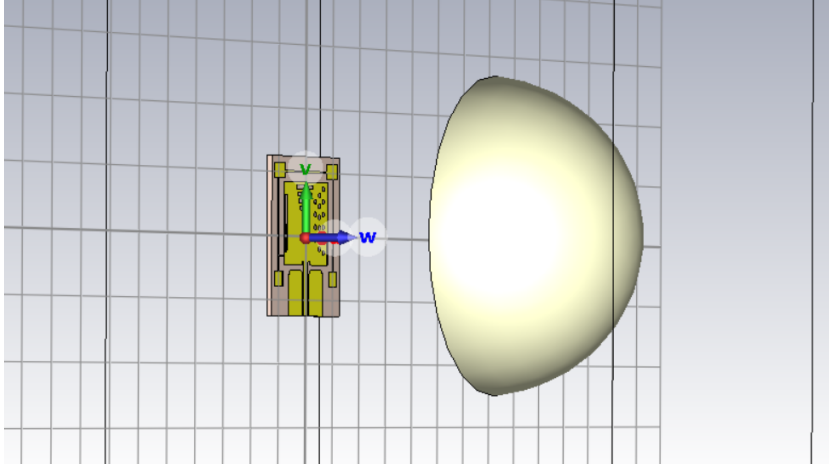


Figure 3.1: CST simulation setup showing the proposed antenna placed in front of the hemispherical breast phantom

3.2 Simulation Setup in CST Studio Suite

The antenna was designed and simulated in CST Studio Suite using the time-domain solver. The simulation was carried out over a frequency range of 0.3 GHz to 5.033 GHz. This frequency range was selected to observe multiple resonant frequencies of the antenna in a single simulation. The antenna was excited using a waveguide port connected to the microstrip feed line, and the S11 values were recorded over the complete frequency range.

This very CST simulation setup was applied later on to perform the breast tumor detection simulation. In this case, the breast phantom, which is hemispherical in shape, was positioned before the antenna, and S11 values were recorded for various positions of the tumor. This dataset was used for the tumor-detection dataset explained in Chapter 4.

3.3 S-Parameter (Return Loss) Results

The simulated S11 response of the proposed antenna is shown in Figure 3.2. The graph shows four clear resonant dips within the 0.3–5 GHz frequency range, which shows the multiband behaviour of the antenna. At each resonant frequency, the S11 value is below the commonly used -10 dB limit. This means that most of the power given to the antenna is accepted by it instead of reflecting it back to the source. The four resonant frequencies and their corresponding S11 values are given in Table 3.2.

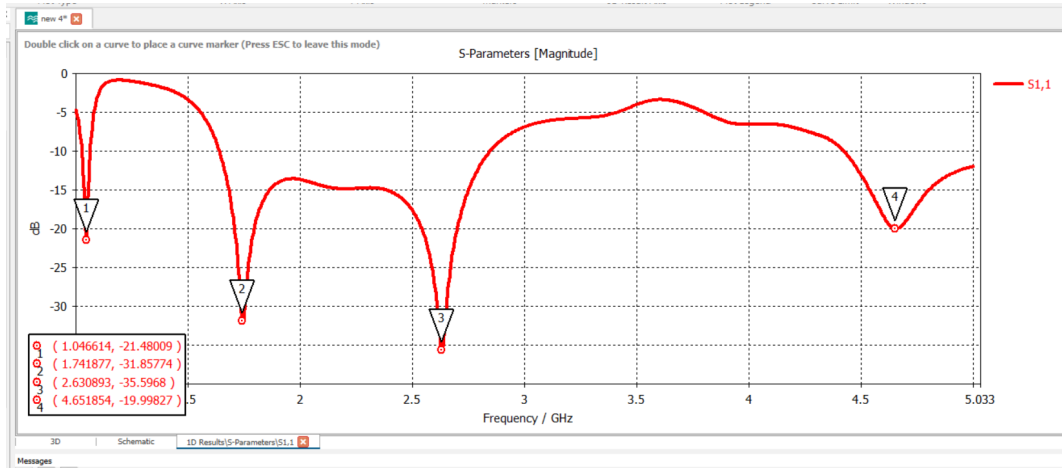


Figure 3.2: Simulated S11 (return loss) response of the proposed antenna, 0.3–5 GHz

Table 3.2: Simulated resonant frequencies and S11 values

Resonance	Frequency (GHz)	S11 (dB)	Remark
Band 1	1.046614	−21.48009	Acceptable matching
Band 2	1.741877	−31.85774	Very good matching
Band 3	2.630893	−35.59680	Best impedance matching
Band 4	4.651854	−19.99827	Acceptable matching; open band*

*See Section 3.4 – the upper edge of Band 4 is not resolved within the simulated frequency range.

Band 3, at 2.630893 GHz, has the lowest S11 value of −35.5968 dB, which means that at this point the antenna has best impedance matching among the four resonant frequencies. All four resonances have S11 values below −10 dB, showing that the antenna has acceptable impedance matching at these frequencies. The resonances are spread over a wide frequency range, from around 1 GHz to 4.65 GHz. This wide range is useful for the proposed antenna because it allows the antenna to operate at different frequencies during the breast tumor sensing simulations.

3.4 Bandwidth Analysis

The impedance bandwidth of the antenna was measured at the −10 dB level. The bandwidth values for the different resonant regions are given in Table 3.3 below.

Table 3.3: Bandwidth of each operating band

Band	Centre Frequency (GHz)	Bandwidth	Bandwidth (MHz)
Band 1	1.046614	0.06 GHz	60
Band 2	1.741877	1.30 GHz	1300
Band 3	2.630893	(spanned by Band 2) [†]	–
Band 4	4.651854	0.60 GHz [‡]	600

[†]Bands 2 and 3 lie close enough in frequency that their -10 dB skirts merge into a single continuous 1.3 GHz wide impedance-matched region spanning both resonances. [‡]Measured bandwidth for the region around Band 4. An important observation from Figure 3.2 is that the fourth resonance at 4.651854 GHz does not fully close within the simulated frequency range. The S_{11} curve goes below -10 dB near 4.65 GHz, but it does not come back above the -10 dB level before the simulation ends at 5.033 GHz. Because of this, the exact upper limit and complete bandwidth of Band 4 cannot be determined from the current simulation. Therefore, a complete bandwidth value is not reported for this band. This limitation can be addressed in future work by increasing the simulation frequency range beyond 5 GHz so that the complete Band 4 response can be observed.

3.5 VSWR Analysis

The Voltage Standing Wave Ratio (VSWR) is directly related to the reflection coefficient $|\Gamma|$, can be calculated from the simulated S_{11} value in dB using the following equation:

$$|\Gamma| = 10^{S_{11}(\text{dB})/20}, \quad \text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (3.1)$$

A VSWR value close to 1 means that the antenna has good impedance matching between the feed line and the patch, which means less power is reflected back to the source. The simulated VSWR response of the antenna is shown in Figure 3.3. The VSWR values at the four resonant frequencies were calculated using Equation 3.1 and the S_{11} values given in Table 3.2. These values are listed in Table 3.4.

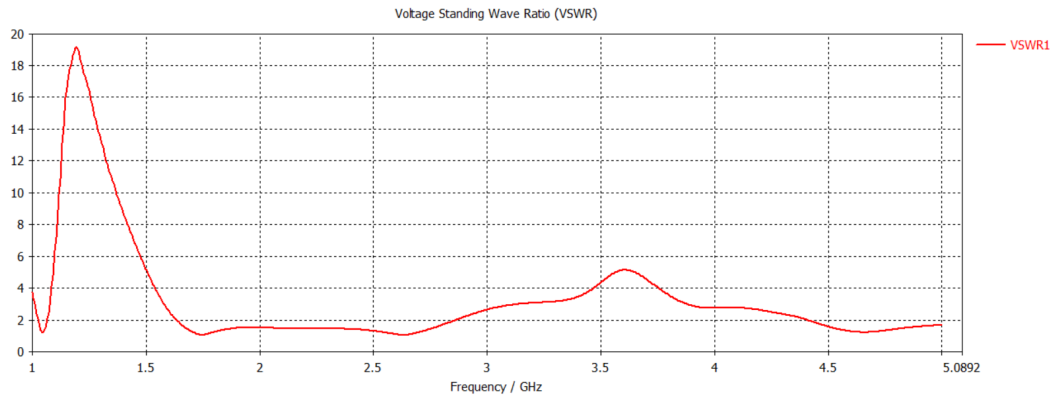


Figure 3.3: Simulated VSWR response of the proposed antenna, 0.3–5 GHz

Table 3.4: VSWR at the resonant frequencies

Resonance	Frequency (GHz)	S11 (dB)	VSWR
Band 1	1.046614	−21.48009	1.18
Band 2	1.741877	−31.85774	1.05
Band 3	2.630893	−35.59680	1.03 (minimum)
Band 4	4.651854	−19.99827	1.22*

*Computed from Equation 3.1 this value was not directly reported in the original CST marker results.

All four resonant frequencies have a VSWR value below 1.25. The first three values, which were directly obtained from the CST markers, are between 1.03 and 1.18. These values are within the commonly used VSWR value of 2 or less for a well-matched antenna. The lowest VSWR value of 1.03 occurs at 2.630893 GHz. This is also the frequency where the lowest S11 value was obtained, which is expected from Equation 3.1.

Between the resonant frequencies, the VSWR increases significantly. For example, a peak of around 19 can be seen near 1.2 GHz in Figure 3.3. This is normal for a multiband antenna because good impedance matching occurs mainly around the resonant frequencies rather than over the complete frequency range.

3.6 Gain, Directivity, and Radiation Pattern

The far-field performance of the antenna was checked at 2.6323 GHz, which is close to the best resonant frequency of Band 3. Figure 3.4 shows the 3D radiation pattern of the antenna at this frequency. From this figure, it can be seen that the radiation is primarily in one direction, which is opposite to the ground plane. For a microstrip patch antenna, this is not surprising at all. This direction of radiation will help in our design because we have placed our antenna facing the breast phantom.

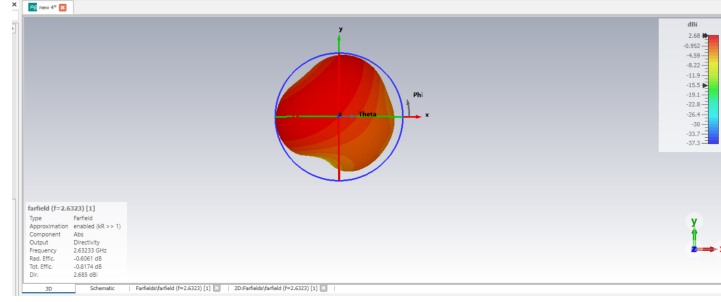


Figure 3.4: Simulated 3-D far-field directivity pattern at 2.63 GHz

The key far-field parameters extracted from CST at this frequency are summarised in Table 3.5.

Table 3.5: Far-field radiation parameters at 2.63 GHz

Parameter	Value
Directivity	2.685 dBi
Radiation efficiency	−0.6061 dB
Total efficiency	−0.8174 dB
Realised gain (simulated)	≈2.076 dBi

Antenna gain, directivity, and radiation efficiency are related by

$$G_{(\text{dB})} = D_{(\text{dB})} + \eta_{\text{rad}, (\text{dB})}, \quad (3.2)$$

Here, D is the directivity, η_{rad} is the radiation efficiency in dB, and G is the gain. Using the values given in Table 3.5 in Equation 3.2, the gain is calculated as

$$[G \approx 2.685 + (-0.6061) = 2.079 \text{ dBi. }]$$

This value is very close to the gain of 2.076 dBi obtained directly from CST. This shows that the calculated value and the CST result are nearly the same. The small difference between the total efficiency of −0.8174 dB and the radiation efficiency of −0.6061 dB is due to some additional loss at the feed port. This is because the antenna was evaluated at 2.6323 GHz, which is slightly different from its best-matched frequency.

3.7 Summary

The proposed 50 mm × 50 mm FR4-based microstrip patch antenna was designed and simulated in CST. The simulation showed four resonant frequencies within the 0.3–5 GHz range. The best matching was obtained at around 2.63 GHz, where the S11 value reached −35.60 dB and the VSWR was 1.03. The antenna also achieved a gain of about 2.08 dBi.

The capability of the multiband antenna makes it work in different frequency bands. This is important in the proposed breast tumor sensing system because the antenna is used in

conjunction with the breast phantom under different tumor scenarios. The output of the simulation of the antenna is used in the proposed tumor-sensing system discussed in the next chapter.

CHAPTER 4

PHANTOM MODELLING AND DATASET ACQUISITION

The present chapter presents the breast phantom applied in the tumor detection simulation experiments. The types of tissues and their dielectric parameters are discussed. Furthermore, variations in the size and location of the tumor applied in each simulation are explained. Finally, the labeled dataset generated by the simulation is presented in the chapter.

4.1 Breast Phantom Design

A three-dimensional hemispherical breast phantom was created in CST Studio Suite and placed in front of the antenna described in Chapter 3, as shown in Figure 3.1. A hemispherical shape is commonly used as a simple model for the breast in microwave sensing studies because it gives a basic representation of the breast shape and is easier to simulate repeatedly [3].

The phantom was made using different tissue layers. From the outside to the inside, it has a thin skin layer, a fat (adipose) layer, and a glandular tissue region. For the tumor-present cases, a small spherical tumor was placed inside the glandular tissue region. The different layers of the phantom are shown in Figure 4.1.

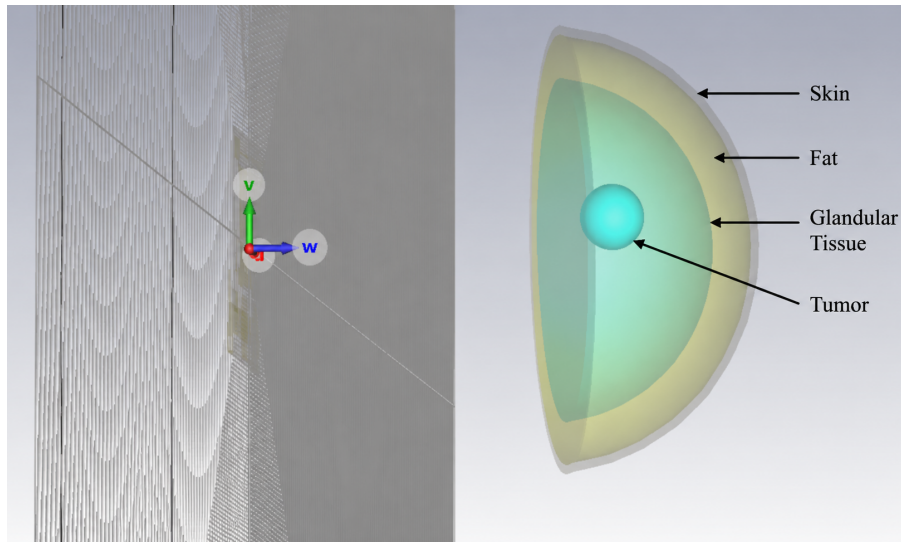


Figure 4.1: Schematic cross-section of the layered hemispherical breast phantom

4.2 Dielectric Properties of Phantom Tissues

Permittivity (ϵ_r), conductivity (σ), and relative permeability (μ_r) values have been given to every tissue layer. These are consistent with the typical values of breast tissue properties found in breast sensing applications.

A high permittivity and conductivity for the tumor tissue compared to fat/glandular tissue around was considered to facilitate microwave detection, as we discussed in Section 2.3. The tissue property values used in the phantom are summarized in Table 4.1.

Table 4.1: Dielectric properties assigned to phantom tissues

Tissue	ϵ_r	σ (S/m)	μ_r
Fat	5.14	0.14	1
Glandular tissue	10.00	0.40	1
Skin	36.00	4.00	1
Tumor	50.00	4.00	1

In this model, the permittivity of the tumor is much higher than that of the fat tissue. The tumor-to-fat permittivity ratio is approximately 9.7:1 (50/5.14). This type of difference in permittivity has also been reported in breast tissue studies [4]. This difference helps the antenna detect changes in the S11 response when a tumor is present.

4.3 Tumor Size and Position Variation

In order to build the dataset, tumors with three diameters, 5 mm, 10 mm, and 15 mm were positioned in the glandular part of the phantom. Different positions for the same tumors were chosen. Since tumor location and diameter were varied, the response of the antennas would be different. Consequently, it makes it possible for machine learning algorithms to learn different cases of tumors rather than just one case.

Moreover, a tumor-free phantom was modeled. This means that the phantom included only the skin, fat, and glandular tissues without any tumors, thus serving as a control case. Different tumor sizes and positions used for modeling are presented in Table 4.2

Table 4.2: Tumor size and position variation summary

Parameter	Description
Tumor diameters modelled	5 mm, 10 mm, 15 mm
Position variation	Tumor translated to multiple distinct positions within the glandular region for each diameter
Total tumor-present configurations	30
Tumor-absent (healthy) configurations	5
Total simulated configurations	35

4.4 Data Acquisition Process

All 35 phantom models were separately simulated in CST Studio Suite, using the same antenna as well as the same simulation parameters mentioned in Chapter 3. 30 of these simulations included tumor of certain size and location, whereas 5 did not have a tumor but included differently positioned phantoms.

The S11 magnitude versus frequency values for each simulation run are stored in a text (.txt) file. All files include two columns – frequency and S11 magnitude. The file names help identify the type of samples. Healthy samples have file names without tumor, tumor samples have file names that include the size of the tumor (either 5mm, 10mm, or 15mm). These names are used when the data is loaded to assign appropriate class label. Class label 0 is assigned to the healthy samples and class label 1 to the tumor-present samples. The whole procedure of data loading and machine learning is described in Chapter 5.

4.5 Dataset Summary

Thus, the total dataset consists of 35 labeled S11 curves, whose summary by class is provided in Table 4.3. The dataset is purposefully imbalanced in favor of tumor-present class (30 out of 35 samples or 85.7%) due to the interest in profiling a variety of tumor sizes and locations. The imbalance is considered in the machine learning technique (Chapter 5).

Table 4.3: Dataset composition summary

Class	Label	No. of Samples	Proportion
Healthy (no tumor)	0	5	14.3%
Cancer (tumor present)	1	30	85.7%
Total	–	35	100%

The 35 labelled S11 curves form the complete raw dataset used in this project. Spectral features were extracted from these S11 curves and then used to train the machine learning models, as explained in the next chapter.

CHAPTER 5

MACHINE LEARNING METHODOLOGY

In this chapter, the machine learning technique that has been applied to the 35 samples of S11 data set which has been explained in Chapter 4 is elaborated upon. This includes feature extraction from the S11 data set, the pre-processing of the data prior to training, various machine learning models that have been applied, evaluation metrics and cross-validation technique employed.

5.1 Overview of the Machine Learning Pipeline

Figure 5.1 shows the complete machine learning process used in this project. Each .txt file obtained from CST contains frequency and S11 magnitude values. First, these values are converted into a fixed set of features for each sample. The complete feature data is then normalised and divided into training and testing data. After that, eight different machine learning models are trained separately. Finally, the performance of all the models is compared using the same evaluation metrics, which are discussed in Chapter 6.

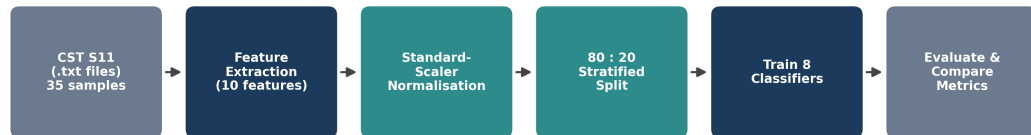


Figure 5.1: Machine learning pipeline from simulated S-parameter data to classifier evaluation

5.2 Feature Extraction from S11 Data

Each simulated S11 curve contains several hundred frequency and magnitude values. These values were converted into a smaller set of ten features for each sample, as shown in Table 5.1. Using a fixed set of features instead of the complete S11 curve makes the data simpler and easier to use for machine learning. This is also useful for this project because the number of samples is small compared to the large number of values present in each S11 curve.

Table 5.1: Extracted S11-domain feature set

Feature	Description
min_s11	Minimum S11 magnitude across the swept band (deepest resonance dip)
max_s11	Maximum S11 magnitude across the swept band
mean_s11	Mean S11 magnitude across the swept band
std_s11	Standard deviation of S11 magnitude across the swept band
median_s11	Median S11 magnitude across the swept band
resonant_freq	Frequency at which the minimum S11 (deepest resonance) occurs
bandwidth_10db	Frequency span between the first and last points where the curve lies more than 10 dB below its own peak level
bandwidth_6db	As above, using a 6 dB threshold
energy	Sum of squared S11 magnitude values across the sweep ($\sum S_{11}^2$)
dynamic_range	Difference between the maximum and minimum S11 magnitude

The features bandwidth_10db and bandwidth_6db were calculated separately for each of the 35 samples using the complete 0.3–5 GHz frequency range. These values should not be confused with the antenna bandwidth given in Section 3.4. In this chapter, they are used as features to describe the shape of each S11 curve. Since the tumor can change the antenna response, these bandwidth values can also change from one sample to another.

5.3 Data Preprocessing

Before training the machine learning models, any invalid values such as NaN or $\pm\infty$ were replaced with zero. After this, all ten features were standardised using scikit-learn’s StandardScaler. This makes the features have a mean of zero and a standard deviation of one. The transformation is given by:

$$z = \frac{x - \mu}{\sigma}, \quad (5.1)$$

Here, μ refers to the mean of the feature, and σ to the standard deviation, both computed based on the training set. Standardization is essential because the range of the value varies in the ten features, e.g., the range of resonant_freq is in units of GHz, whereas energy is the summation of squared values. If the features are not standardized, features with large values might play a greater role in the model. In particular, KNN and SVM models would

be influenced significantly by the magnitude of the features.

In order to standardize the dataset, an 80:20 train:test split was conducted. Since the ratio of samples with and without tumors needs to be maintained, stratified splitting (`stratify=y`) was applied here. Using `random state=42` makes sure that the same train:test split is used every time the code is run. The final training and test samples can be found in Table 5.2.

Table 5.2: Train-test split summary

Partition	Healthy	Cancer	Total
Training set (80%)	4	24	28
Test set (20%)	1	6	7
Full dataset	5	30	35

5.4 Machine Learning Models

Eight supervised machine learning algorithms were trained and tested using the standardised features. A short explanation of each model and the hyperparameters used are given below and summarised in Table 5.3.

5.4.1 Logistic Regression

Logistic Regression is a type of machine learning algorithm which is applied for classification purposes. Using Logistic Regression, the algorithm calculates the probability of a data point being in one specific class based on its features. This algorithm was used in this study as the simplest one to compare other machine learning algorithms. Equal weighting of classes accounted for the imbalance of the two classes in the dataset.

5.4.2 K-Nearest Neighbours (KNN)

KNN [13] is a machine learning algorithm where classification of a new sample depends on the classes of its closest samples within the dataset. In our case, the closest samples were found based on their Euclidean distance from the test sample after the features standardisation. The parameter `k` had the value of 5, meaning that the classes of the five closest samples are taken into account when a new sample is classified. The KNN algorithm does not learn a separate model. Instead, it memorises the training data.

5.4.3 Support Vector Machine (SVM)

SVM [14] is an algorithm that helps to classify two classes by trying to find a margin which would provide the maximum gap between two classes. In the current project, the RBF kernel is applied to SVM algorithm. This makes it possible for the algorithm to deal with nonlinear

dependencies in the dataset and draw an appropriate margin between two classes. Moreover, class weights are balanced due to the uneven distribution of samples in each class. The probability estimates option was also enabled in order to calculate the ROC-AUC value.

5.4.4 Decision Tree

The Decision Tree is a decision-making algorithm that splits the dataset into smaller datasets according to the values of the features. For every step, the model chooses an appropriate feature and its value to split the dataset into multiple subsets. In this project, the maximum depth of the tree was defined as 10. The main advantage of Decision Trees is that they are interpretable, but the disadvantage is overfitting, which may happen due to the relatively small size of the dataset.

5.4.5 Random Forest

Random Forest [15] is a machine learning model that uses many Decision Trees together to make a prediction. In this project, 100 decision trees were used. Each tree was trained using a different part of the training data and a random selection of features. The final prediction is made by combining the predictions of all the trees. This helps reduce overfitting and gives more stable results than using only one Decision Tree. Class weights were balanced, and the maximum depth of each tree was set to 10.

5.4.6 XGBoost

XGBoost [16] is a machine learning model that uses multiple Decision Trees one after another. Each new tree tries to improve the mistakes made by the previous trees. In this project, 100 boosting rounds were used. The maximum depth of each tree was set to 5, and the learning rate was set to 0.1.

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5.4.9 *Random Forest*

The Random Forest [15] algorithm belongs to the family of machine learning models using ensembles of Decision Trees for prediction. In this work, the number of decision trees used was equal to 100. Every tree was trained with the help of different parts of the train set and different features. The final decision is obtained by aggregation of decisions of all the trees, which leads to better stability and less overfitting compared to using just one Decision Tree. The balance of class weights was set to True and the maximum tree depth was equal to 10.

5.4.10 *XGBoost*

XGBoost [16] is an algorithm that belongs to machine learning models, which use gradient boosting and multiple Decision Trees sequentially. Every tree is trained to minimize errors of all the previous trees. In this work, the number of boosting rounds used was equal to 100. Maximal tree depth was equal to 5 and the learning rate was equal to 0.1.

5.4.11 *CatBoost*

CatBoost [17] is yet another algorithm from the family of machine learning models using gradient boosting and multiple Decision Trees sequentially. CatBoost also uses ordered boosting, which decreases the effect of overfitting, especially for small datasets. In this work, the number of iterations used was equal to 100. Tree depth was 5 and the learning rate of 0.1.

5.4.12 *Artificial Neural Network (ANN)*

An Artificial Neural Network (ANN) was implemented using Keras for binary classification. It used hidden layers of 64, 32, and 16 neurons with ReLU activation and dropout of 0.3. A sigmoid output was used for classification. The model was trained using Adam and binary cross-entropy for 50 epochs with a batch size of 4.

Table 5.3: Hyperparameters of the eight classifiers

Classifier	Key Hyperparameters
Logistic Regression	max_iter=1000, class_weight='balanced'
KNN	n_neighbors=5 (Euclidean distance)
SVM	kernel='rbf', probability=True, class_weight='balanced'
Decision Tree	max_depth=10
Random Forest	n_estimators=100, max_depth=10, class_weight='balanced'
XGBoost	n_estimators=100, max_depth=5, learning_rate=0.1
CatBoost	iterations=100, depth=5, learning_rate=0.1
ANN	64–32–16–1 dense layers, dropout 0.3, Adam optimiser, 50 epochs, batch size 4

The classifiers used random_state=42 wherever it was applicable. This helped keep the results consistent each time the models were run.

5.5 Evaluation Metrics

All the classifiers were tested using the same test set given in Table 5.2. Six different evaluation metrics were used to compare the models. These metrics were calculated from the confusion matrix, which contains True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) values. In this project, the positive class represents “Cancer”, meaning that a tumor is present.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5.2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5.3)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (5.4)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5.5)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5.6)$$

For the Tumor Classification System, Recall is extremely critical since it is an indication of how many true tumor cases have been found by the algorithm. Low Recall may

indicate that some cases of tumors may not be detected. Precision refers to how many out of those classified as a tumor actually are tumors, thus helping in identifying the false positives. F1-score is a measure of balance between precision and recall. Specificity refers to how well the algorithm distinguishes healthy cases from tumor cases. This is also critical in this project since there are few healthy cases as in Table 5.2.

In addition to other metrics of evaluation, the ROC-AUC metric was also calculated for all the classifiers. The difference between the ROC-AUC and the accuracy or F1-score is that ROC-AUC is not based on one particular value of the decision threshold. Instead, it assesses the ability of the classifier to correctly discriminate between tumor and normal instances using different decision thresholds. The higher the ROC-AUC, the better the discrimination ability of the classifier.

Lastly, the time necessary for training and prediction of each classifier was also measured in seconds. This data may be important if the system is used for rapid diagnosis of tumors in the future.

5.6 Cross-Validation Strategy

Other than splitting the dataset in an 80:20 ratio between train and test, 5-fold stratified cross-validation was applied to verify the model performance. The process involved dividing the 35 samples into 5 sets. Each time, one of the sets was set aside for testing while the remaining four sets were used for training. The process was done 5 times such that all 5 sets got to be used for testing. Accuracy, F1-score, and ROC-AUC values were recorded for all models.

5.7 Implementation Details

The complete machine learning work was done in Python. Models such as Logistic Regression, KNN, SVM, Decision Tree, and Random Forest were implemented using `scikit-learn`. XGBoost and CatBoost were used through their Python packages, and the ANN was implemented using TensorFlow/Keras. The performance of the models was calculated using `scikit-learn.metrics`. The best model, along with the scaler and feature information, was saved using `joblib` so that it can be used again in future work.

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CHAPTER 6

RESULTS AND DISCUSSION

In this chapter, the eight machine learning algorithms that were utilized in this research are outlined. The performance of the algorithms is compared in terms of their performance and time taken to complete the task. Also included is an analysis of the confusion matrices of the algorithms. Reasons for using KNN on this dataset are discussed.

6.1 Model Performance Comparison

Table 6.1 shows the Accuracy, Precision, Recall, Specificity, F1-Score, and ROC-AUC values of all eight machine learning models. The models were tested using the 7-sample test set mentioned in Table 5.2. Figure 6.1 compares the Accuracy, Precision, Recall, and F1-Score of all eight models.

Table 6.1: Performance comparison of the eight classifiers

Model	Accuracy	Precision	Recall	Specificity	F1-Score	ROC-AUC
Logistic Regression	85.71%	100.00%	83.33%	100.00%	90.91%	1.000
KNN	100.00%	100.00%	100.00%	100.00%	100.00%	1.000
SVM	85.71%	85.71%	100.00%	0.00%	92.31%	1.000
Decision Tree	100.00%	100.00%	100.00%	100.00%	100.00%	1.000
Random Forest	85.71%	100.00%	83.33%	100.00%	90.91%	1.000
XGBoost	85.71%	100.00%	83.33%	100.00%	90.91%	1.000
CatBoost	100.00%	100.00%	100.00%	100.00%	100.00%	1.000
Artificial Neural Network	85.71%	100.00%	83.33%	100.00%	90.91%	1.000

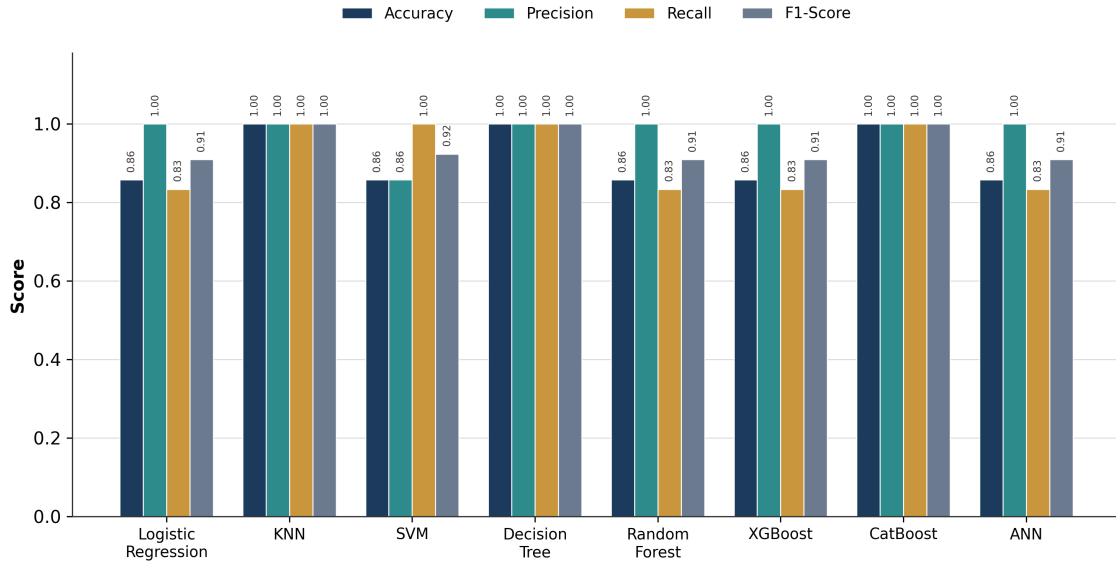


Figure 6.1: Comparative performance (Accuracy, Precision, Recall, F1-Score) of the eight classifiers

Three of the eight models –KNN,Decision Tree,and CatBoost – achieved 100% accuracy, precision, recall, specificity, and F1-score on the test set.The other five models – Logistic Regression, SVM, Random Forest, XGBoost, and ANN – achieved 85.71% accuracy.However,the errors made by these models were different, as discussed in Section 6.2.All eight models also achieved a ROC-AUC of 1.000,showing that the models were able to separate the two classes well even though some models made one incorrect prediction at the default 0.5 threshold.

6.2 Confusion Matrix Analysis

Figure 6.2shows the confusion matrix of KNN on the 7-sample test set, which contains 6 Cancer samples and 1 Healthy sample.KNN correctly classified all 6 Cancer samples and the 1 Healthy sample.Therefore, there were no false positives or false negatives.

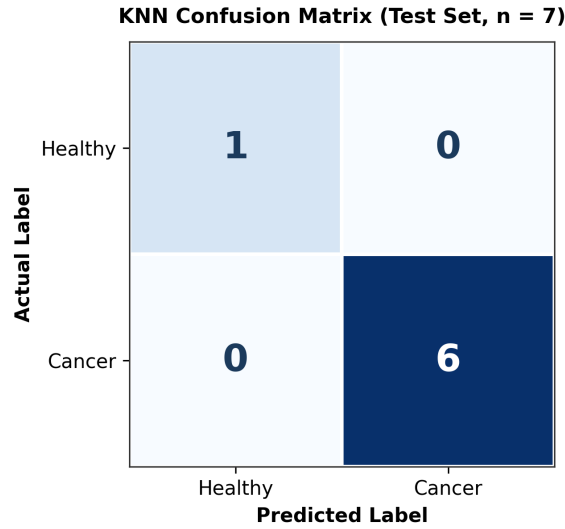


Figure 6.2: Confusion matrix of the best-performing model (KNN) on the test set

Table 6.2 shows the confusion matrix values for all eight models. These values were obtained from the Precision, Recall, and Specificity results given in Table 6.1.

Table 6.2: Confusion matrix summary for all classifiers (test set, $n = 7$)

Model	TP	TN	FP	FN
Logistic Regression	5	1	0	1
KNN	6	1	0	0
SVM	6	0	1	0
Decision Tree	6	1	0	0
Random Forest	5	1	0	1
XGBoost	5	1	0	1
CatBoost	6	1	0	0
Artificial Neural Network	5	1	0	1

In the non-accurate models, there are two kinds of errors. In Logistic Regression, Random Forest, XGBoost, and ANN model, one sample from class with tumor was misclassified into the healthy class. In such way, we got one false negative with the recall equal to 83.33%, while the healthy sample was classified properly.

The SVM model had an opposite error. It properly classified all six samples with tumor presence but failed to recognize the one healthy sample and classified it as a tumor. So, we have 100% recall but no specificity. For cancer screening, the omission of tumor detection is usually more important than false positive. But the fact is that there was only one healthy sample in the testing dataset.

All eight models had the same ROC-AUC value - 1.000, although some of them made one error in their prediction. This means that these models were able to make the right

ranking of samples from two classes based on their predicted probability. The mistakes were made in cases when probabilities of samples were close to the 0.5 boundary.

6.3 Computational Efficiency

Table 6.3 and Figure 6.3 compare training and prediction time across the eight classifiers.

Table 6.3: Computational cost comparison

Model	Training Time (s)	Prediction Time (s)
Logistic Regression	0.0089	<0.001
KNN	<0.001	0.0069
SVM	0.0074	<0.001
Decision Tree	0.0020	0.0013
Random Forest	0.1085	0.0159
XGBoost	0.0562	0.0020
CatBoost	0.3215	<0.001
Artificial Neural Network	8.0637	0.1203

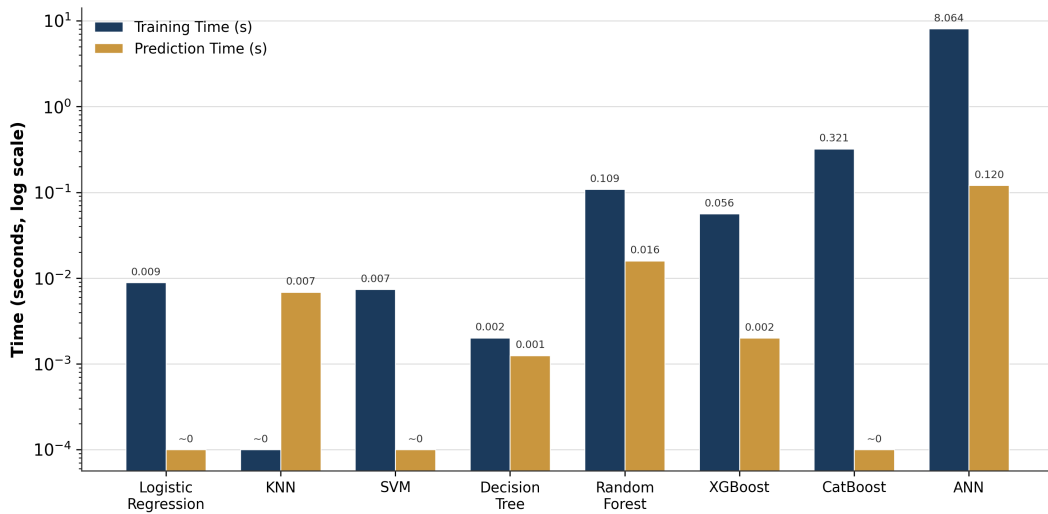


Figure 6.3: Training and prediction time comparison across the eight classifiers (log scale)

The training phase for ANN was the most time-consuming, taking 8.06 s. This is far higher than the other models where CatBoost took just 0.32 s. ANN is time-consuming during the training process since it undergoes several iterations, which is 50 in our case.

KNN model does not involve any significant amount of training time since it stores the training data but does not create any model. All the computations for KNN happen during the prediction when it tries to find the distance between the training data points and the new point. KNN took 0.0069 s for prediction in our test.

6.4 Why K-Nearest Neighbours Was Identified as the Best Model

KNN, Decision Tree, and CatBoost all achieved 100% on the main evaluation metrics shown in Table 6.1. Among these three models, KNN was considered the most practical choice for this project for the following reasons.

- **Very low training time.** As shown in Table 6.3, KNN takes almost no time to train. Decision Tree takes 0.0020 s, while CatBoost takes 0.3215 s. Even though all three models gave the same test results, KNN required much less training time.
- **Simple to understand.** KNN is easy to understand because it makes a prediction by looking at the closest labelled samples. This makes it easier to explain compared to more complex models.
- **Works well with the small dataset.** Only 28 samples were used for training with ten features. Since the dataset is small, KNN is a suitable choice because it can work well when the two classes are clearly separated in the feature space.

However, it is worth mentioning that Decision Tree and CatBoost have also demonstrated similar results on the test set. Thus, KNN should not be seen as the optimal model in all situations. The selection of KNN is mostly justified by the simplicity of its architecture and fast training time in case of the particular data sample. As the test sample includes only 7 examples, one should consider the results obtained carefully, as stated in Section 6.6.

6.5 Note on Cross-Validation

Furthermore, as mentioned in Section 5.6, 5-fold stratified cross-validation was done to evaluate the performance of the models on the complete 35-sample dataset. However, the exact results of the analysis have not been shown in this report. They could be obtained easily by running the pipeline script from the Appendix. In this case, comparison of the results obtained from one test set and from the cross-validation would be helpful.

6.6 Limitations

While the results obtained in this study are encouraging, there are several limitations that need to be considered when interpreting the findings.

1. **Very small dataset.** The study uses only 35 samples, including 5 healthy samples. Since only 1 healthy sample was present in the test set, the results can change significantly with even one incorrect prediction. Therefore, the 100% scores should be treated as a promising initial result rather than a final diagnostic result.
2. **Simulation-only validation.** All results were obtained from CST simulations and were not tested using a physical antenna or real measurements. Real breast tissue is more com-

plex than the simplified phantom used in this study and can also introduce measurement noise and changes in antenna position.

3. **Risk of overfitting.** Models such as Decision Tree and KNN can perform very well on small datasets but may not give the same results on new data. Their performance should therefore be checked using a larger dataset and further validation.
4. **Limited antenna-phantom configurations.** The simulations used fixed antenna positions. The effect of changing the antenna position or direction was not studied.
5. **Unresolved fourth resonant band.** The upper limit of the fourth resonance at around 4.65 GHz could not be fully determined because the simulation ended at 5.033 GHz. Therefore, its complete bandwidth and contribution to the machine learning features could not be fully studied.

These limitations directly motivate the future work directions outlined in Chapter 7.

6.7 Summary

Out of the total eight models, three models, namely KNN, Decision Tree, and CatBoost models, gave a perfect 100% accuracy in testing the 7 samples used for the test phase. Other five models provided an accuracy rate of 85.71%. In this regard, KNN was deemed to be the most practical because it provided a perfect result in the test phase, simple architecture, and minimal training time.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The current paper presents the procedure of design, simulation, and machine learning of microwave antenna system for breast tumor detection purposes. The study has been carried out during one-month internship period at National Institute of Technology, Jamshedpur. Three main topics have been researched within the scope of the project which include microstrip antenna design, layering of breast phantom for simulation, and classification of results with machine learning algorithms.

A microstrip patch antenna with size $50 \text{ mm} \times 50 \text{ mm}$ has been designed and simulated with lossy FR4 substrate material in CST Studio Suite in the frequency range of 0.3–5 GHz. The results indicate that there are four resonance frequencies of the antenna which are 1.05, 1.74, 2.63, and 4.65 GHz. The best matching has been observed at the frequency of 2.63 GHz at which the value of S11 equals to -35.60 dB, and the VSWR is equal to 1.03. The simulated gain is 2.08 dBi. The results have been also verified by using the relationship between gain, directivity, and efficiency discussed in Section 3.6.

Breast phantom with hemispherical form has been designed in CST software and 35 sets of S11 data has been generated which includes 30 cases with breast with tumors and 5 cases with healthy breast. 5 mm, 10 mm and 15 mm tumors have been considered. Ten features have been extracted from S11 data and evaluated with eight machine learning algorithms. KNN, Decision Tree, and CatBoost demonstrated 100% test accuracy and KNN is the most simple among them.

In conclusion, it can be stated that the combination of low cost microwave antenna with machine learning algorithms could help to detect whether there is a tumor in the breast phantoms in a simulation environment. However, there is the lack of data, especially healthy data. The results can be considered as preliminary research rather than clinically validated system.

7.2 Future Scope

Based on the work done in this project, the following improvements can be considered in future work:

1. **Fabrication of hardware and measurements.** The proposed antenna can be fabricated on an FR4 substrate and then tested by means of a vector network analyzer. It will

help in comparison of results of the real antenna with simulations done in CST. Tissue mimicking phantoms may be used for additional measurements.

2. **Expanding the dataset.** It is possible to increase the amount of training data by adding information about different sizes of tumors, their locations, and also healthy samples. The addition of more healthy samples will increase reliability of machine learning results and decrease the effect of individual samples on the result of the tests, as described in Section 6.6.
3. **Complete characterization of the fourth resonant band.** The simulation frequency range should be expanded up to 5 GHz in order to completely characterize the fourth resonant band. It will help to determine its upper bandwidth limit and check whether it can be used as an additional band.
4. **Physical verification of the machine learning pipeline.** In the future, the ML models should be trained and tested with real S11 measurements instead of only simulation results. It will be necessary to find out whether the same set of features and machine learning models works for real measurements.
5. **Investigation of image and deep-learning techniques.** In the future, the raw S-parameter data should be provided directly as input for deep learning models such as 1D CNNs or RNNs. The investigation of large antenna arrays can be done in order to localize the tumor, not just detecting it.
6. **Robustness of the antenna location.** In the future, different phantom setups can be tested where antenna's position and orientation vary. It will be necessary to investigate performance of the ML models in this case.
7. **Clinical validation.** In the future, the proposed system should be tested with patient data. It will be necessary to collaborate with doctors in order to do so.

In this way, the combination of an improved antenna and machine learning algorithm employed in this project may be further enhanced to become a relatively inexpensive and portable solution that does not use ionization.

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